

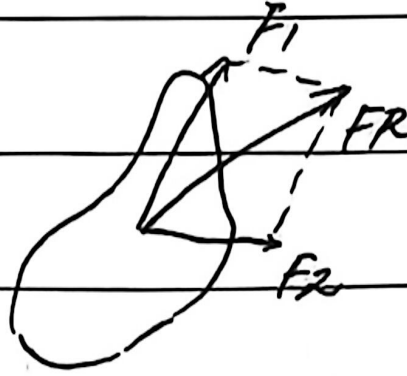
Ch1. 静力学基本公理

物体的受力分析

§1.1 静力学基本公理

公理1. 平行四边形法则

$$\vec{F_R} = \vec{F_1} + \vec{F_2}$$



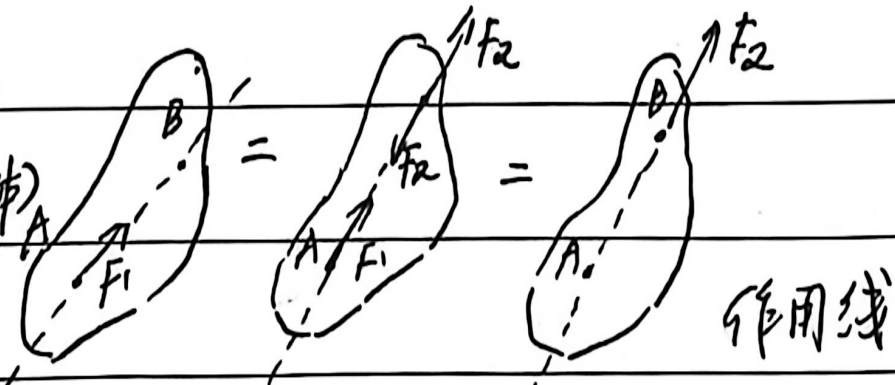
三角形法则

公理2 (加减平衡力系公理) 二力平衡条件

公理3. 加减平衡力系公理

推论1. 力的可传性 (刚体)

沿着力的作用线平移力。

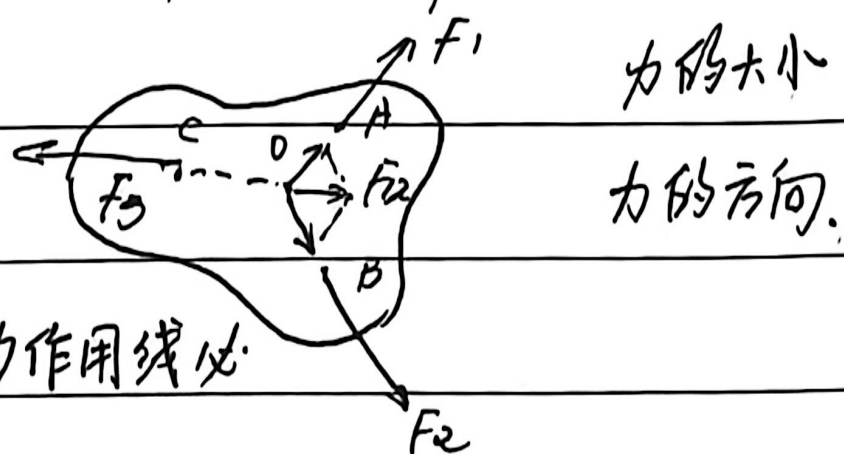


推论2. 三力平衡汇交定理。

三个力作用于刚体并平衡

若二力可以汇交于一点 则第三力作用线必

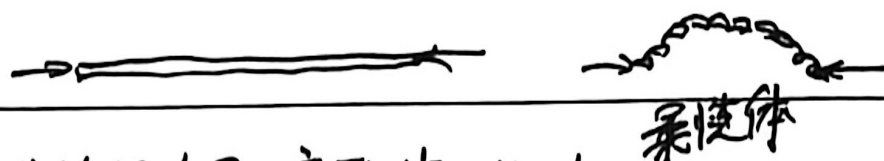
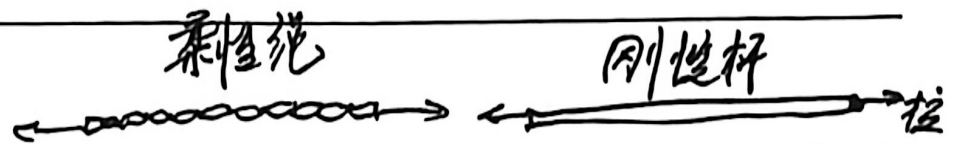
过此点, 且在同一平面。



公理4. 作用和反作用定律

公理5 刚化公理。

柔性体在平衡条件下较化成刚体, 其平衡状态不变。反之不成立。



二力平衡, 加减平衡力系力的可传性不适用于变形体以及三力汇交。

主动力 $\Rightarrow$ 被动力 约束力

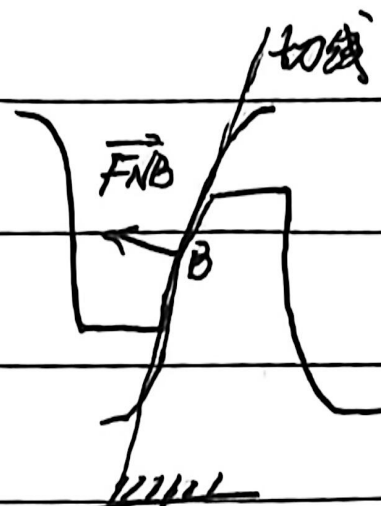
§1.2 约束和约束力

限制物体运动的其它物体. 靠力来约束.

约束 $\rightarrow$ 约束力

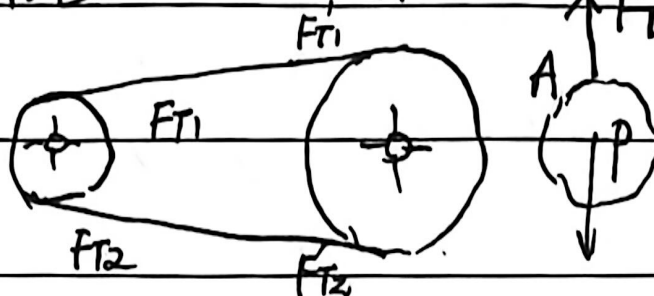
1. 光滑接触约束. (无摩擦) ($\vec{F}_N$)

过接触点 (垂直) 公法线方向指向被约束物体



2. 柔索类约束 ($\vec{F}_T$)

过接触点或过绳切割处 沿着绳子轴线方向背离被约束物体.



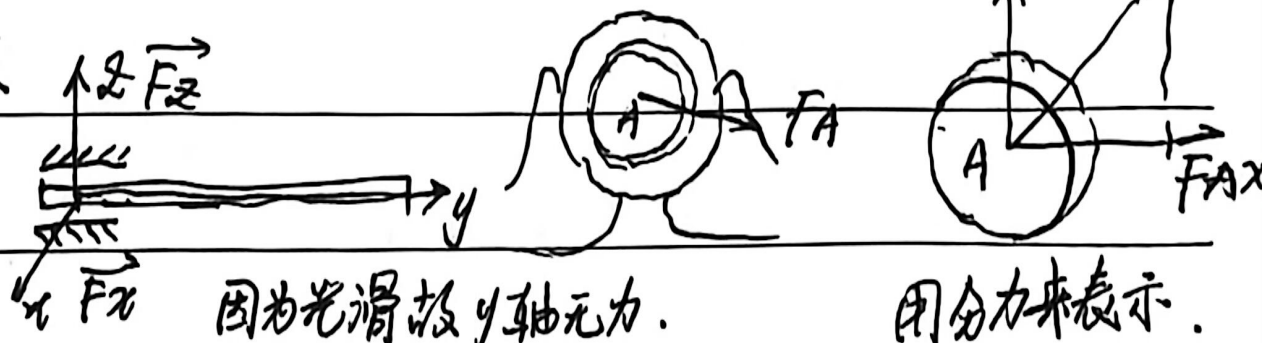
3 光滑圆柱铰链约束.

(1) 向心轴承

F_{T1} 与 F_{T2} 非作用力与反作用力

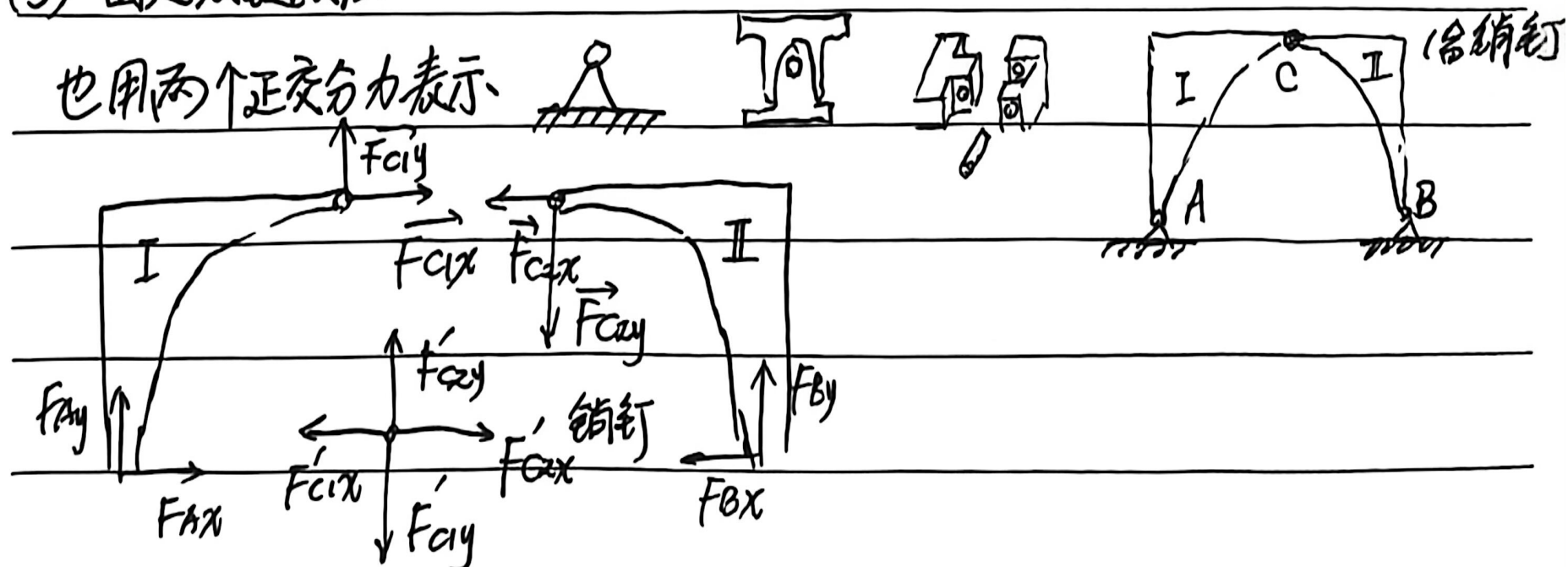
定义同光滑接触约束

(2) 光滑圆柱铰链.



(3) 固定铰链支座

也用两个正交分力表示.



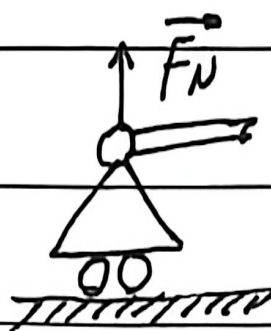
连接多个物体将销钉单独拿出来分析.

画受力图时只需类型对即可无需规定方向

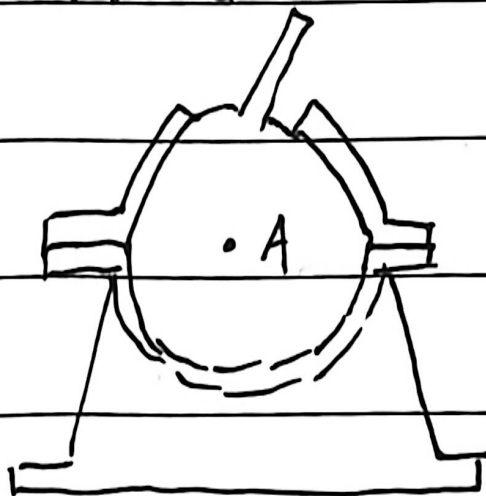
4 其他约束

1. 可动铰链支座

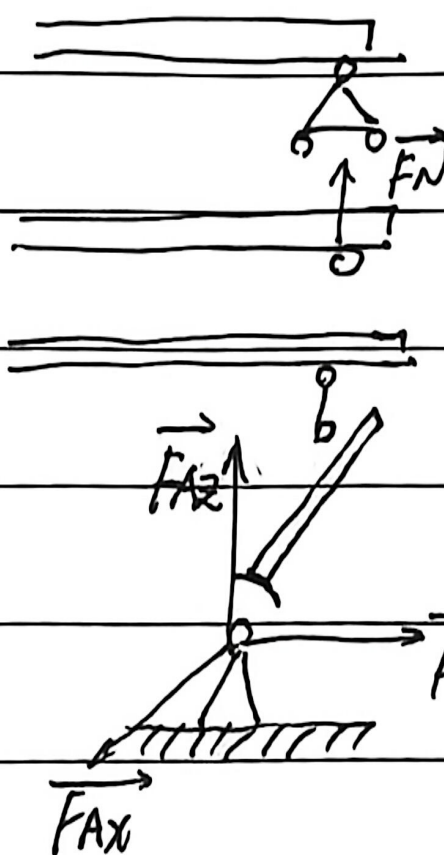
水平方向可移动, 无约束力



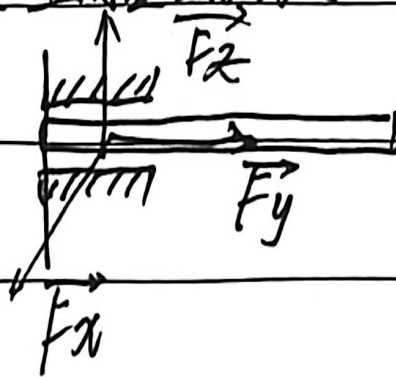
2. 球铰链 (光滑)



也属于光滑接触约束



3. 止推轴承

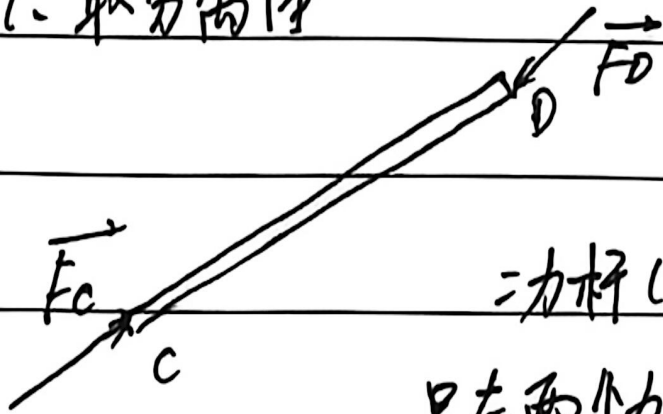


§1.3 物体的受力

分析和画受力图

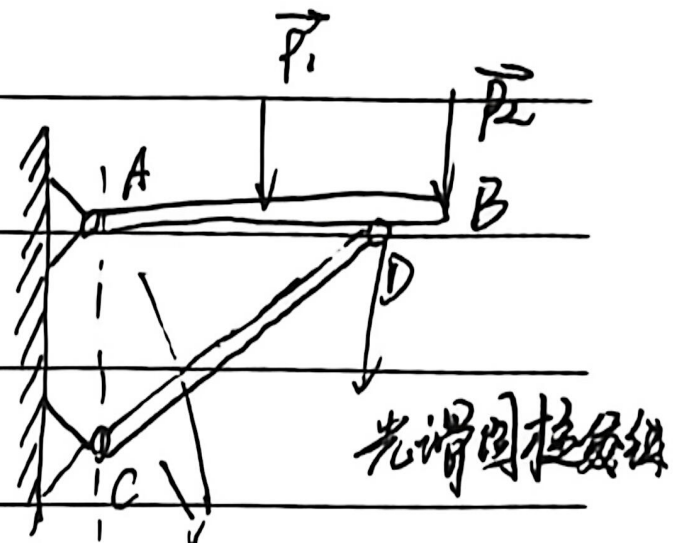
例1. P_1 杆自身重量.

1. 取分离体



二力杆 (二力构件)

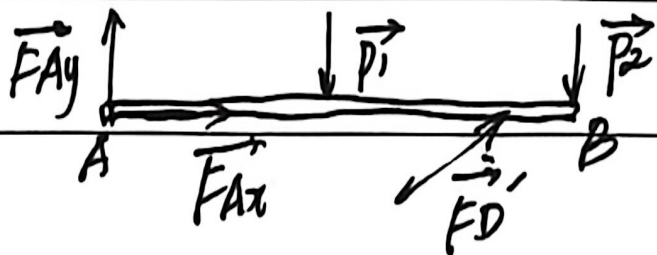
只在两个力作用下平衡的构件.



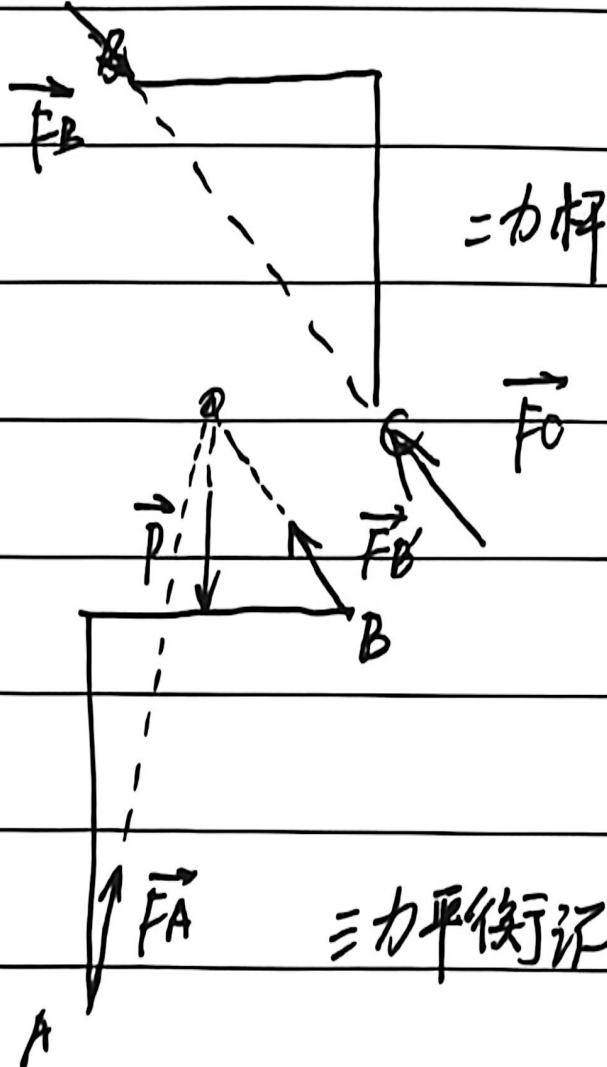
固定铰链支座

作用线沿两作用点连线方向

2. 先画主动力后画约束力



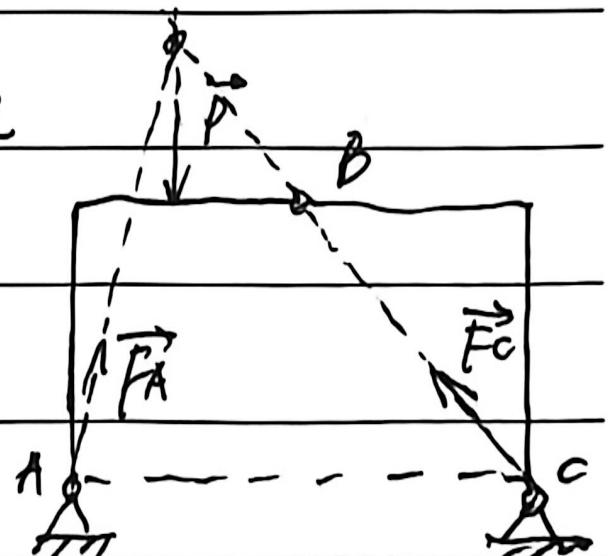
3 作用力与反作用力



二力杆

三力平衡汇交定理 (找方向)

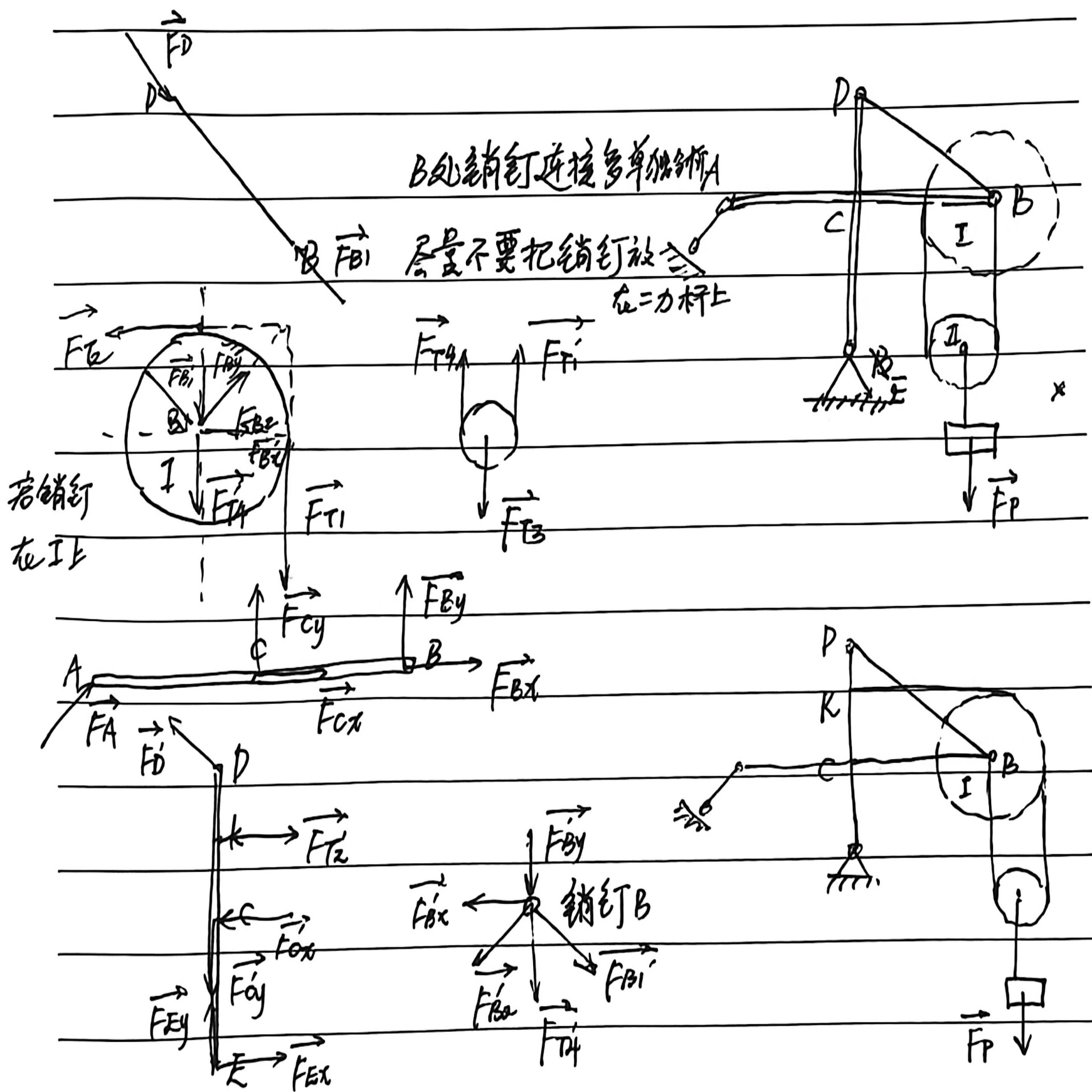
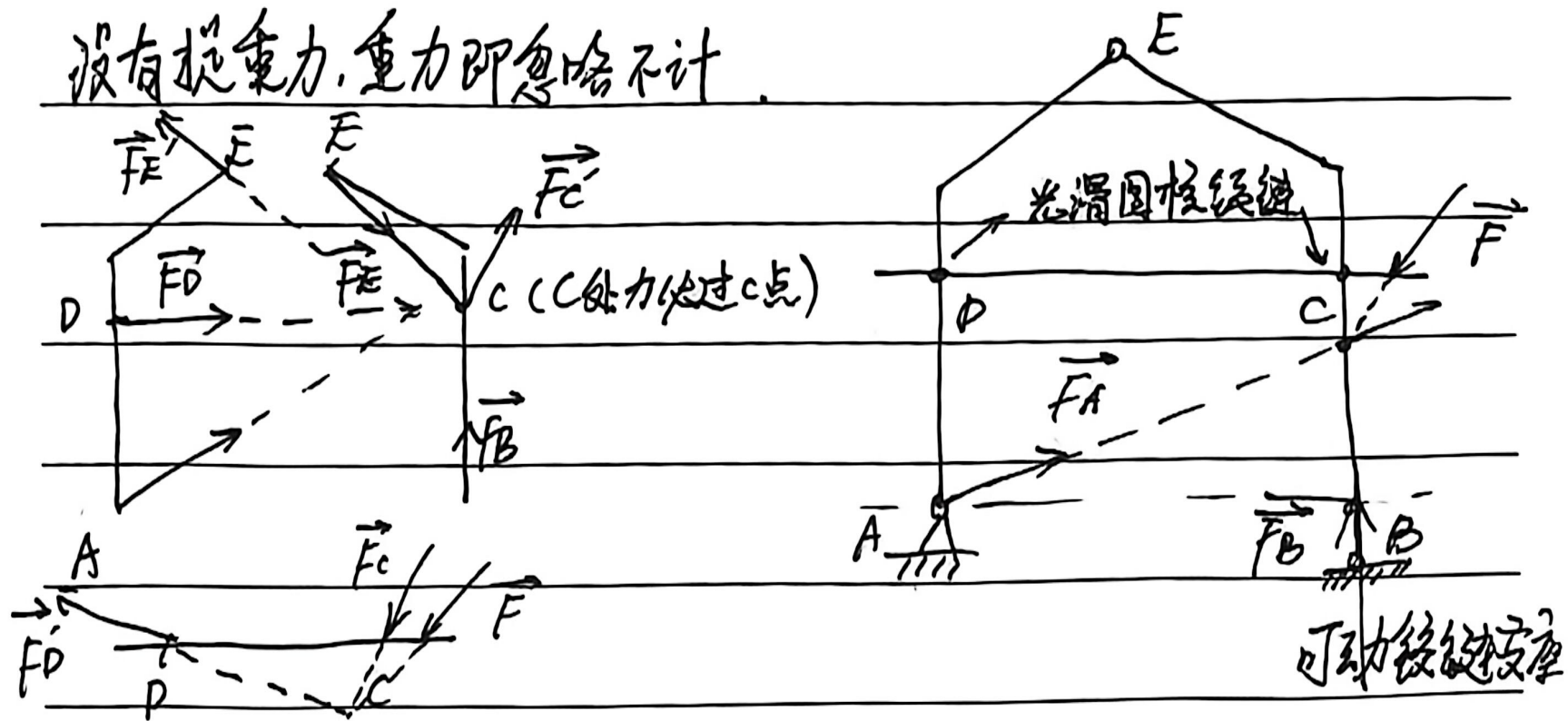
例2

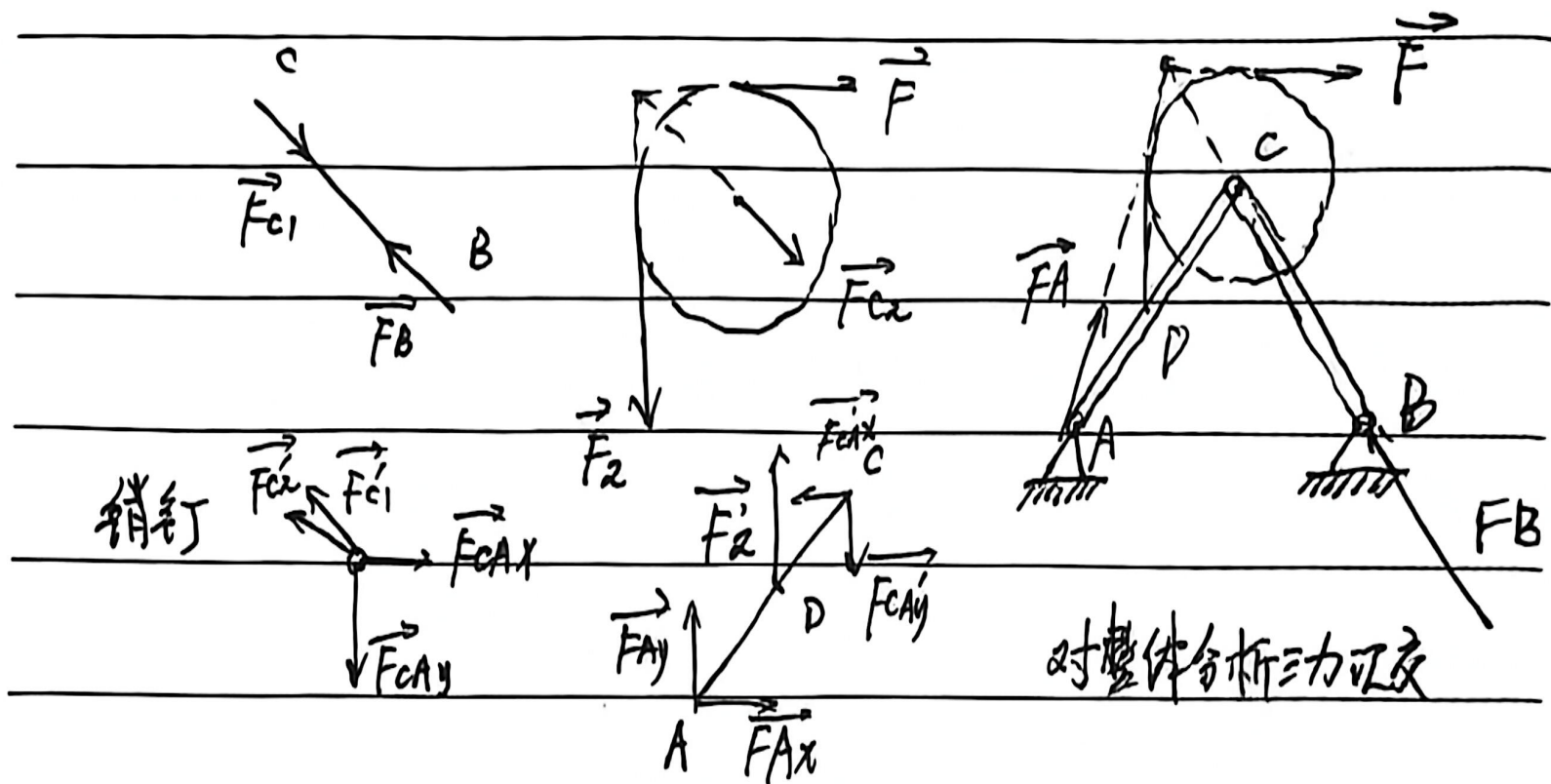


4 内力和外力的转换

例3.

没有提重力,重力即忽略不计.





§1.4 力学模型

忽略变形 $\Rightarrow$ 刚体

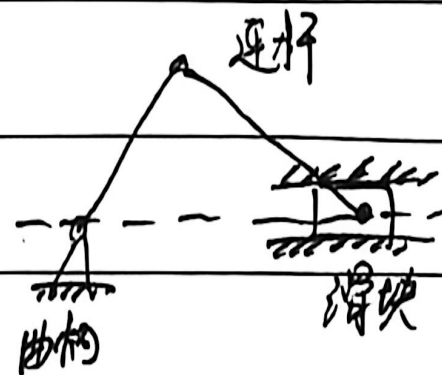
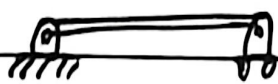
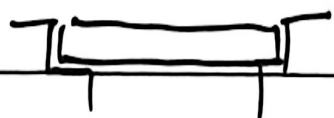
三维问题 $\Rightarrow$ 平面问题

几何形状 $\Rightarrow$ 圆形

力的简化 $\Rightarrow$ 作用在圆心

AB处约束力简化 $\Rightarrow$ 点接触光滑接触。

简支梁



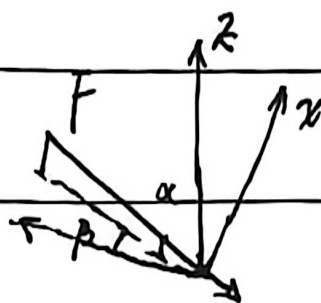
§2.1 汇交力系

一. 空间力的投影

$$\begin{array}{lcl}
 \vec{F}_x = \vec{F}_1 \cos \varphi & \left. \begin{array}{l} \\ \\ \end{array} \right\} & \begin{array}{l} \text{一} \\ \text{次} \\ \text{投} \\ \text{影} \\ \text{法} \end{array} \\
 \vec{F}_y = \vec{F}_1 \cos \theta & & \\
 \vec{F}_z = \vec{F}_1 \cos \psi & &
 \end{array}
 \qquad
 \begin{array}{lcl}
 \vec{F}_x = \vec{F}_1 \sin \tau \cos \varphi & \left. \begin{array}{l} \\ \\ \end{array} \right\} & \begin{array}{l} \text{二} \\ \text{次} \\ \text{投} \\ \text{影} \\ \text{法} \end{array} \\
 \vec{F}_y = \vec{F}_1 \sin \tau \sin \varphi & & \\
 \vec{F}_z = \vec{F}_1 \cos \tau & &
 \end{array}$$

$$\begin{array}{l}
 \vec{F}_x = \vec{F}_n \cos \alpha \sin \beta \times (-1) \\
 \vec{F}_y = \vec{F}_n \cos \alpha \cos \beta \times (-1) \\
 \vec{F}_z = \vec{F}_n \sin \alpha \times (-1)
 \end{array}$$

不带负号

二 简化(合成) $\Rightarrow$ 一个力

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2$$

$$\Rightarrow \vec{F}_R = \sum_{i=1}^n \vec{F}_i = \sum \vec{F}_i \quad (1. \text{几何法, 力的多边形封闭边})$$

2 解析法.

合矢量投影定理.

$$\begin{array}{lcl}
 F_{Rx} = \sum F_{ix} \\
 F_{Ry} = \sum F_{iy} \\
 F_{Rz} = \sum F_{iz}
 \end{array}
 \Rightarrow F_R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2 + (\sum F_z)^2}$$

$$\cos(\vec{F}_R, \vec{i}) = \frac{F_{Rx}}{F_R} = \frac{\sum F_{ix}}{F_R}$$

三 平衡

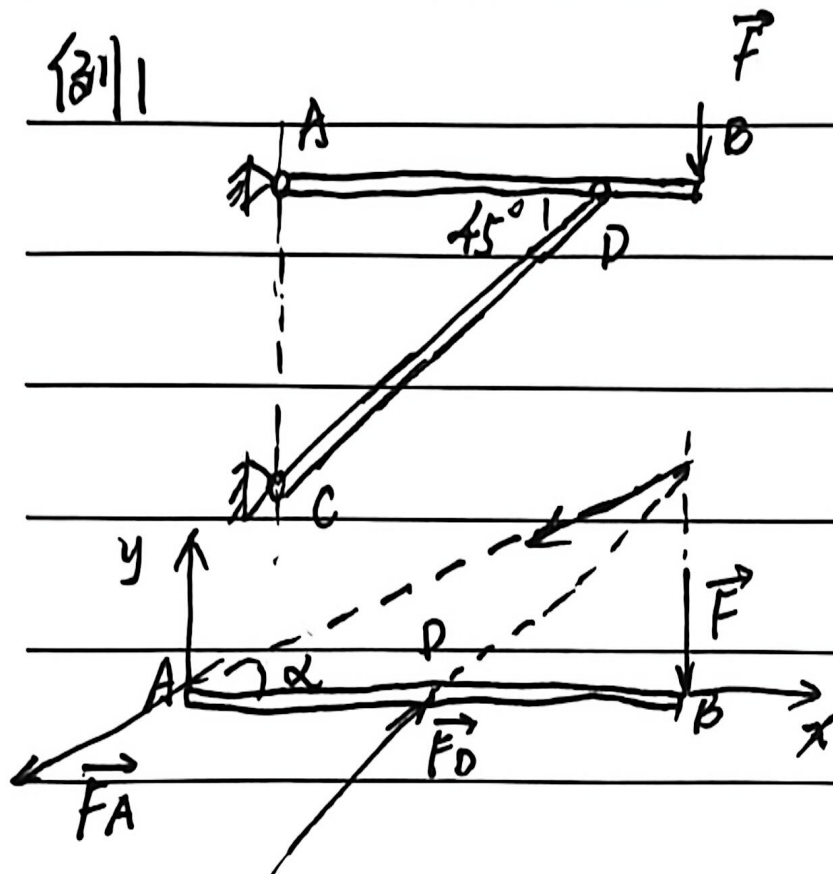
1. 几何法 $\vec{F}_R = \vec{0}$, 力的多边形自行封闭2 解析法 $\sum F_{ix} = 0$ $\sum F_{iy} = 0$ 平衡方程. $\sum F_{iz} = 0$

平面汇交力系

$$\sum F_{\Sigma x} = 0$$

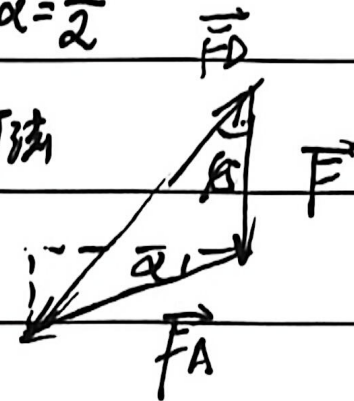
$$\sum F_{\Sigma y} = 0$$

例1

已知 $F = 10 \text{ kN}$ $AD = BD$ 求 A 和 B 处的反力

$$\tan \alpha = \frac{1}{2}$$

几何法

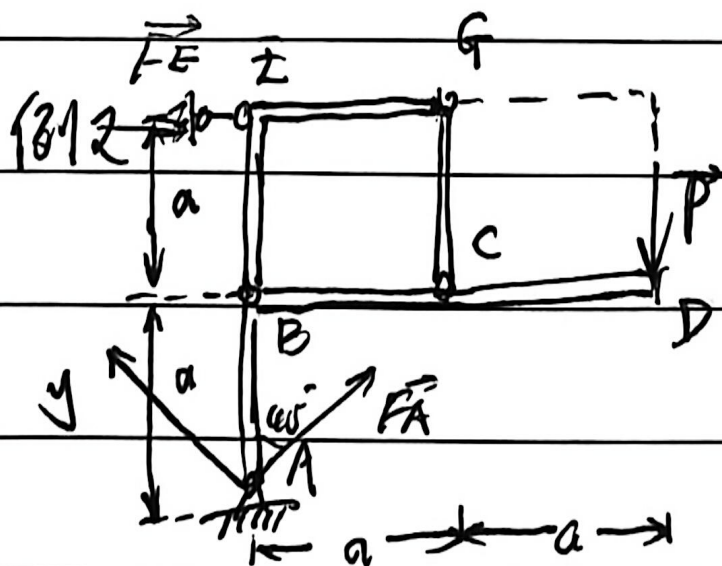


解析法

$$\sum F_{\Sigma x} = 0 \quad -F_A \cos \alpha + F_D \cos 45^\circ = 0$$

$$\sum F_{\Sigma y} = 0 \quad -F + F_D \sin 45^\circ - F_A \sin \alpha = 0$$

$$\Rightarrow F_A, F_D$$



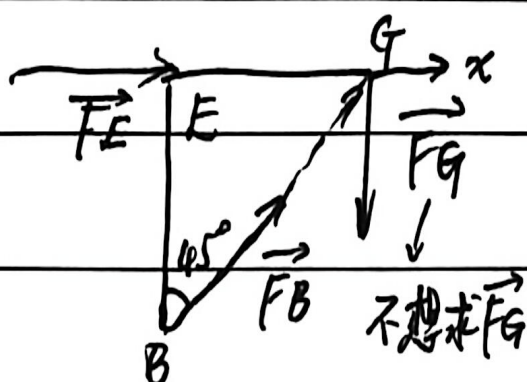
整体

三力平衡

$$\sum F_y = 0 \quad -P \cos 45^\circ - F_E \cos 45^\circ = 0$$

$$\Rightarrow F_E = -P$$

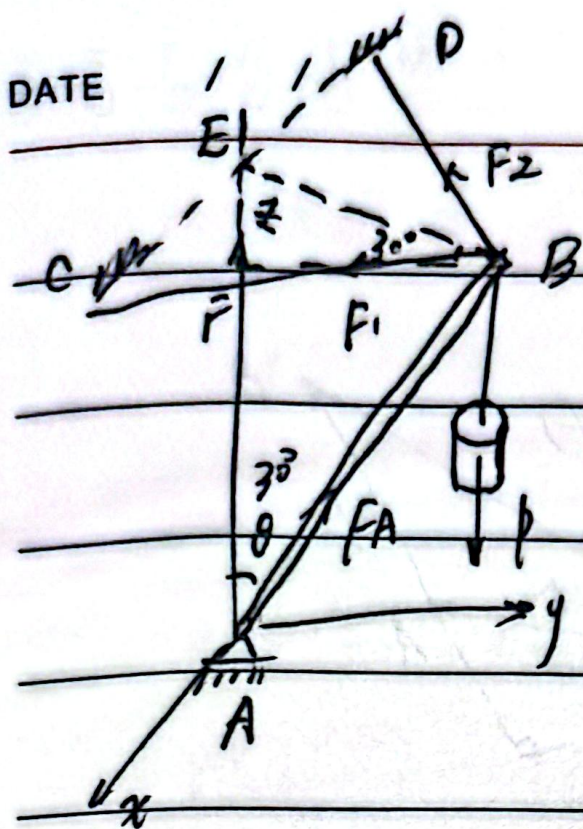
可动铰链支座可以承受拉力



$$\sum F_x = 0 \quad F_E + F_B \sin 45^\circ = 0$$

$$F_B = \sqrt{2}P$$

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$$P = 10 \text{ kN} \quad CE = EB = DE \quad \theta = 30^\circ$$

B点

$$\sum F_{\Sigma x} = 0.$$

$$F_1 \sin 45^\circ - F_2 \sin 45^\circ = 0.$$

$$\sum F_{\Sigma y} = 0.$$

$$-(F_1 + F_2) \cos 45^\circ \cos 30^\circ + F_A \sin 30^\circ = 0.$$

二次投影.

$$\sum F_{\Sigma z} = 0$$

$$(F_1 + F_2) \cos 45^\circ \sin 30^\circ + F_A \cos \theta - P = 0.$$

解 F_1, F_2, F_A

§2.2 力对点的矩和力对轴的矩

一、平面力对点的矩

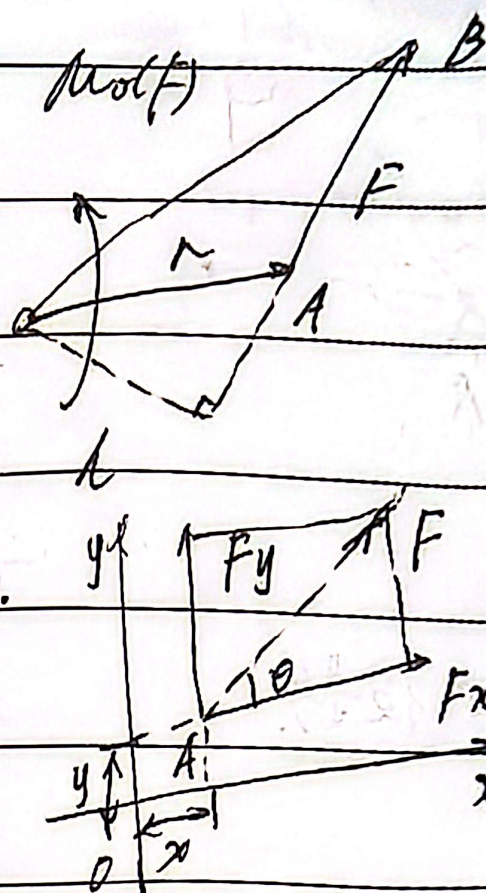
$$M_o(\vec{F}) = +Fh \quad (\text{逆转为正})$$

二、合力矩定理

$$M_o(\vec{F}) = M_o(\vec{F}_x) + M_o(\vec{F}_y)$$

$$M_o(\vec{F}) = -F_x y + F_y x \quad \text{解析式}$$

以矩心为原点建立坐标系



三、空间力对点的矩

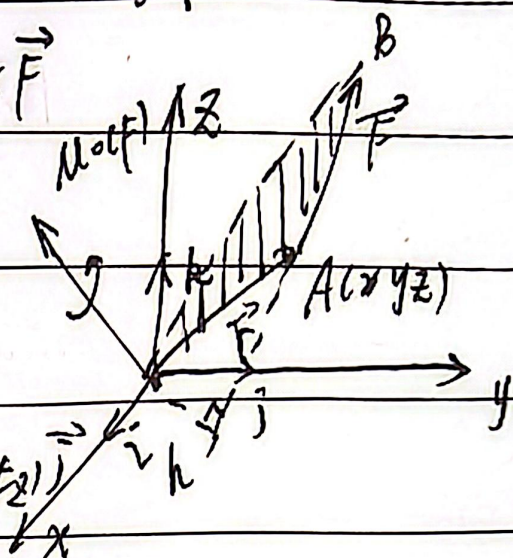
$$\text{力矩作用面方位} \Rightarrow \vec{M}_o(\vec{F}) = \vec{r} \times \vec{F}$$

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\Rightarrow \vec{M}_o(\vec{F}) = (yF_z - zF_y)\vec{i} + (zF_x - xF_z)\vec{j} + (xF_y - yF_x)\vec{k}$$

$$\vec{M}_o(\vec{F}) = \sum \vec{M}_o(\vec{F}_i) \quad \text{合力矩定理}$$

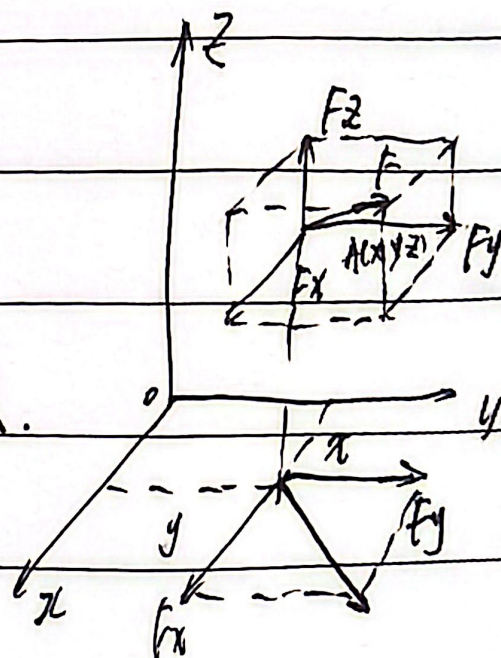


四、力对轴的矩

$$M_z(\vec{F}) = M_o(\vec{F}_{xy}) = -F_x y + F_y x$$

$$M_z(\vec{F}) = [M_o(\vec{F})]_z \quad (\text{投影})$$

五、力对点的矩和力对轴的矩的关系



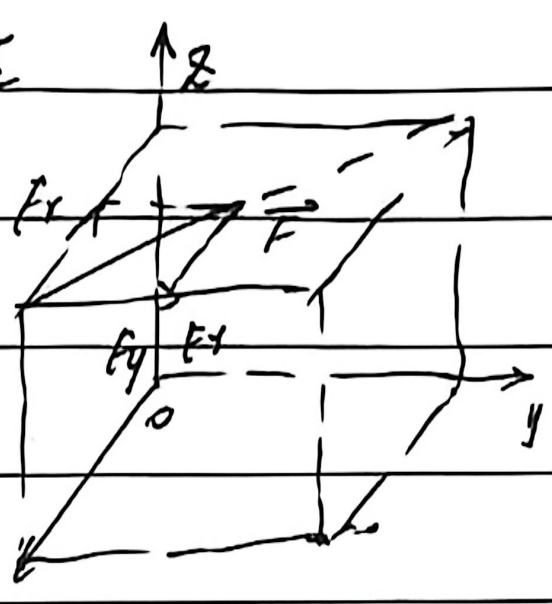
例1 正立方体 a $\vec{F}$ 求对 x, y, z 及 O 点的矩

$$\begin{aligned} M_x(\vec{F}) &= M_x(\vec{F}_x) + M_y(\vec{F}_y) \\ &= -F \cos 45^\circ a. \end{aligned}$$

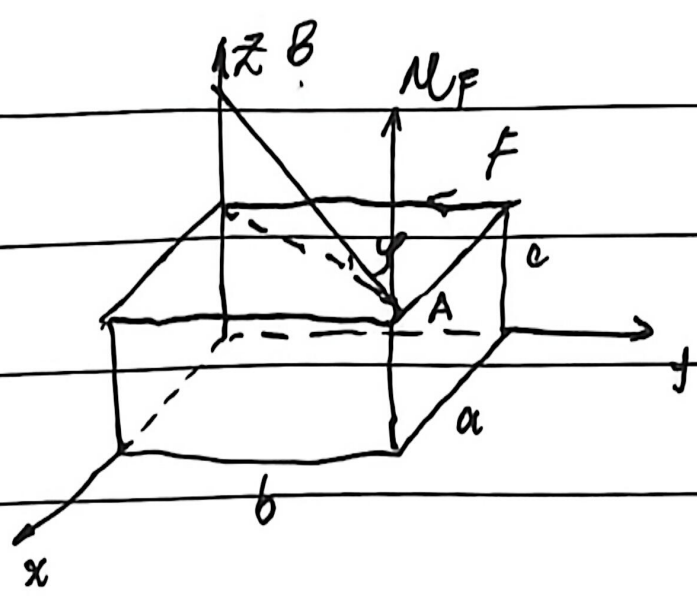
$$M_y(\vec{F}) = M_y(\vec{F}_x) = -F \cos 45^\circ a.$$

$$M_z(\vec{F}) = M_z(\vec{F}_y) = F \cos 45^\circ a$$

$$M_O(\vec{F}) = -F \cos 45^\circ a \vec{i} - F \cos 45^\circ a \vec{j} + F \cos 45^\circ a \vec{k}$$

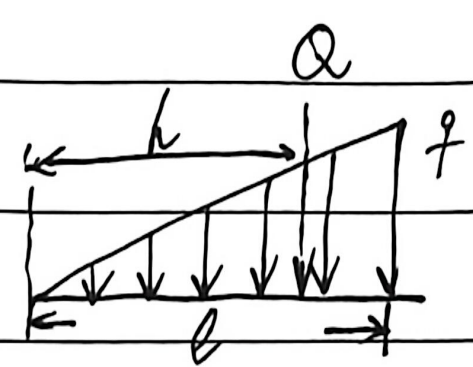
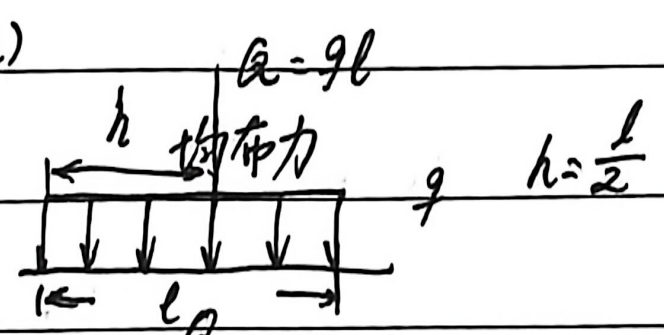


已知力 F 及长方体的边长 a, b, c 则力对 AB 轴的矩为 $F \cos \alpha \cdot c$ AB 轴 A 指向 B

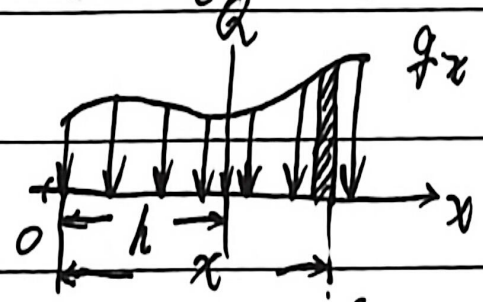


分布力 { 均布力 $q = \text{常量}$
非均布力

$\nearrow \text{KN/m (N/m)}$



$$q_x = \frac{q_0}{l} x$$



$$Q = \int_0^l q_x dx$$

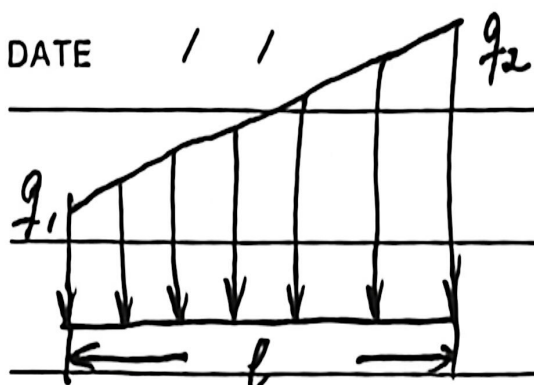
$$Q = \frac{1}{2} q l$$

$$Q \cdot h = \int_0^l q_x \cdot x dx$$

$$h = \frac{2}{3} l$$

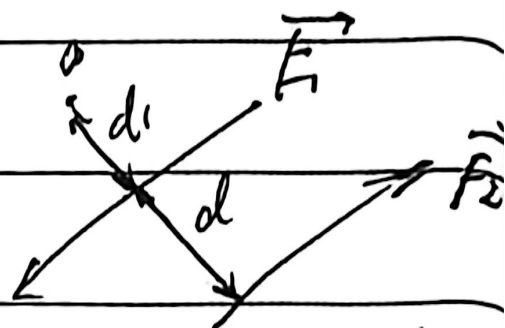
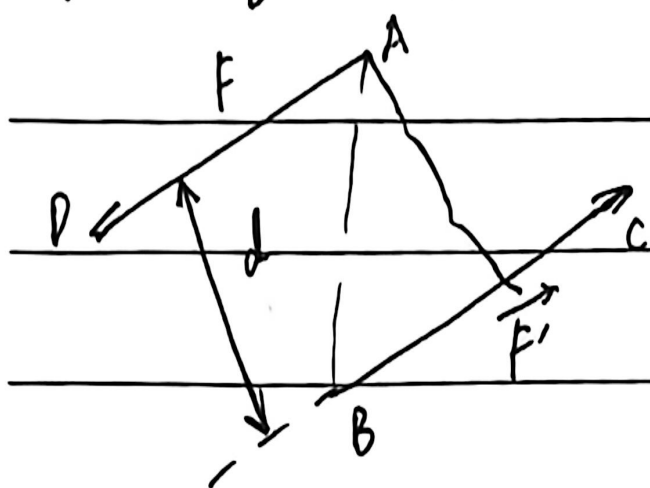
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§ 2.3 力偶理论

一、力偶矩



$$M_O(\vec{F}_1, \vec{F}_2) = M_O(\vec{F}_1) + M_O(\vec{F}_2) \quad F_1 =$$

$$= -F_1 d_1 + F_2 (d + d_1) = F d \Rightarrow M$$

$$M = \pm F \cdot d \text{ 平面问题}$$

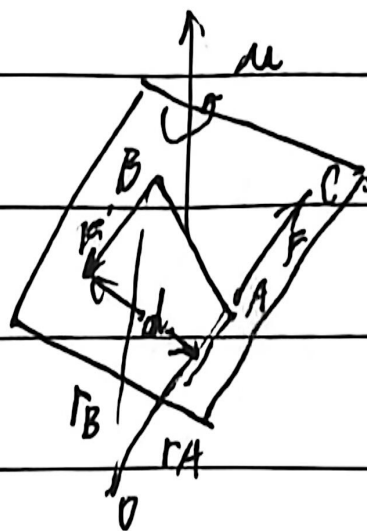
$$\Rightarrow \vec{M}$$

空间力偶 力偶的作用面

$$\vec{M} = \vec{r}_A \times \vec{F} + \vec{r}_B \times \vec{F}$$

$$= (\vec{r}_A - \vec{r}_B) \times \vec{F}$$

$$\Rightarrow \vec{M} = \vec{r}_{BA} \times \vec{F} \text{ 自由矢量}$$



2. 力偶的等效定理.

力偶相等 $\Rightarrow$ 推论
 M $\int M$ 为偶矩
三、力偶系的简化 $\Rightarrow$ 一个力偶

$$\vec{M} = \sum \vec{M}_i \Rightarrow \sum M_{ix} = M_x \Rightarrow M = \sqrt{(M_{ix})^2 + (M_{iy})^2 + (M_{iz})^2}$$

$$\sum M_{iy} = M_y \quad \cos(\vec{M}, \vec{i}) = \frac{\sum M_{ix}}{M}$$

$$\sum M_{iz} = M_z$$

$$M_x = \sum M_{ix} = -M_3 - (M_4 + M_5) \cos 45^\circ$$

$$M_y = \sum M_{iy} = -M_2$$

$$M_z = \sum M_{iz} = -M_1 - (M_4 + M_5) \sin 45^\circ$$

四力偶系的平衡

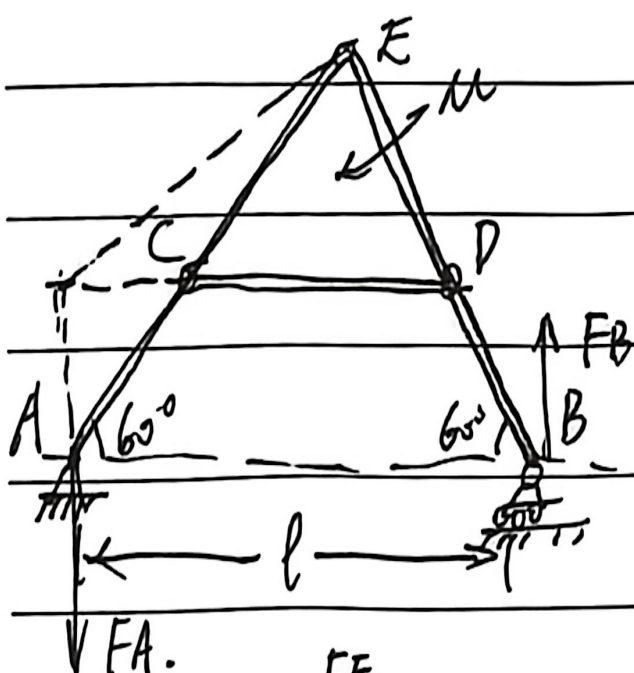
 $\vec{R}=0 \Rightarrow \sum M_{ix}=0$ 空间力偶系的平衡方程.

$$\sum M_{iy}=0$$

$$\sum M_{iz}=0$$

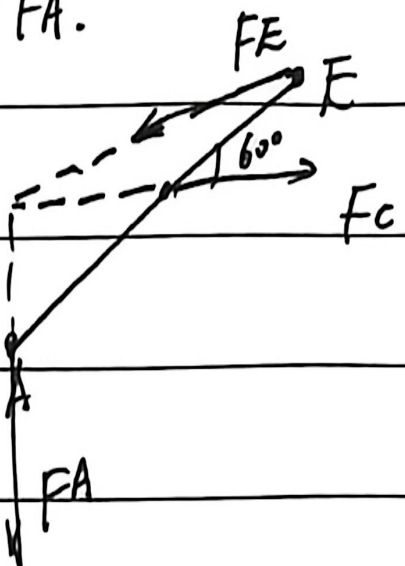
$$\sum M_i=0$$

平面力偶系的平衡方程.

例1 已知 M , CD 为中点求 AE 处约束力整体 $F_A = F_B$

$$\sum M_i = 0. -M + F_A l = 0$$

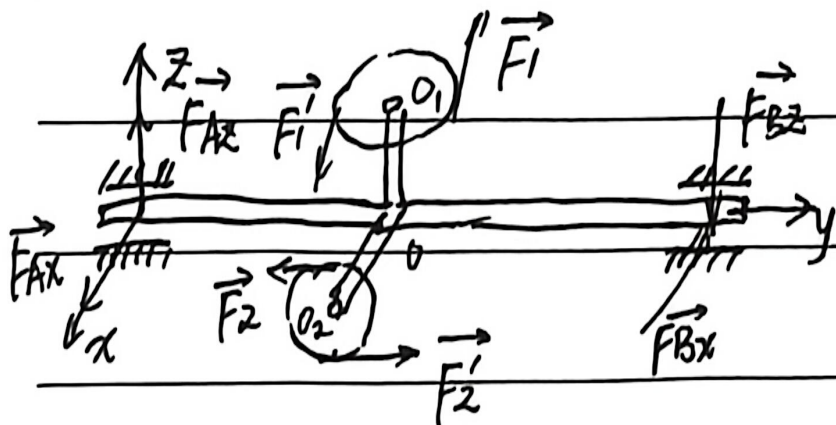
$$F_A = \frac{M}{l}$$



$$\sum F_{iy} = 0$$

$$-F_A - F_E \sin \alpha = 0$$

$$\Rightarrow F_E = -\frac{F_A}{\sin \alpha}$$

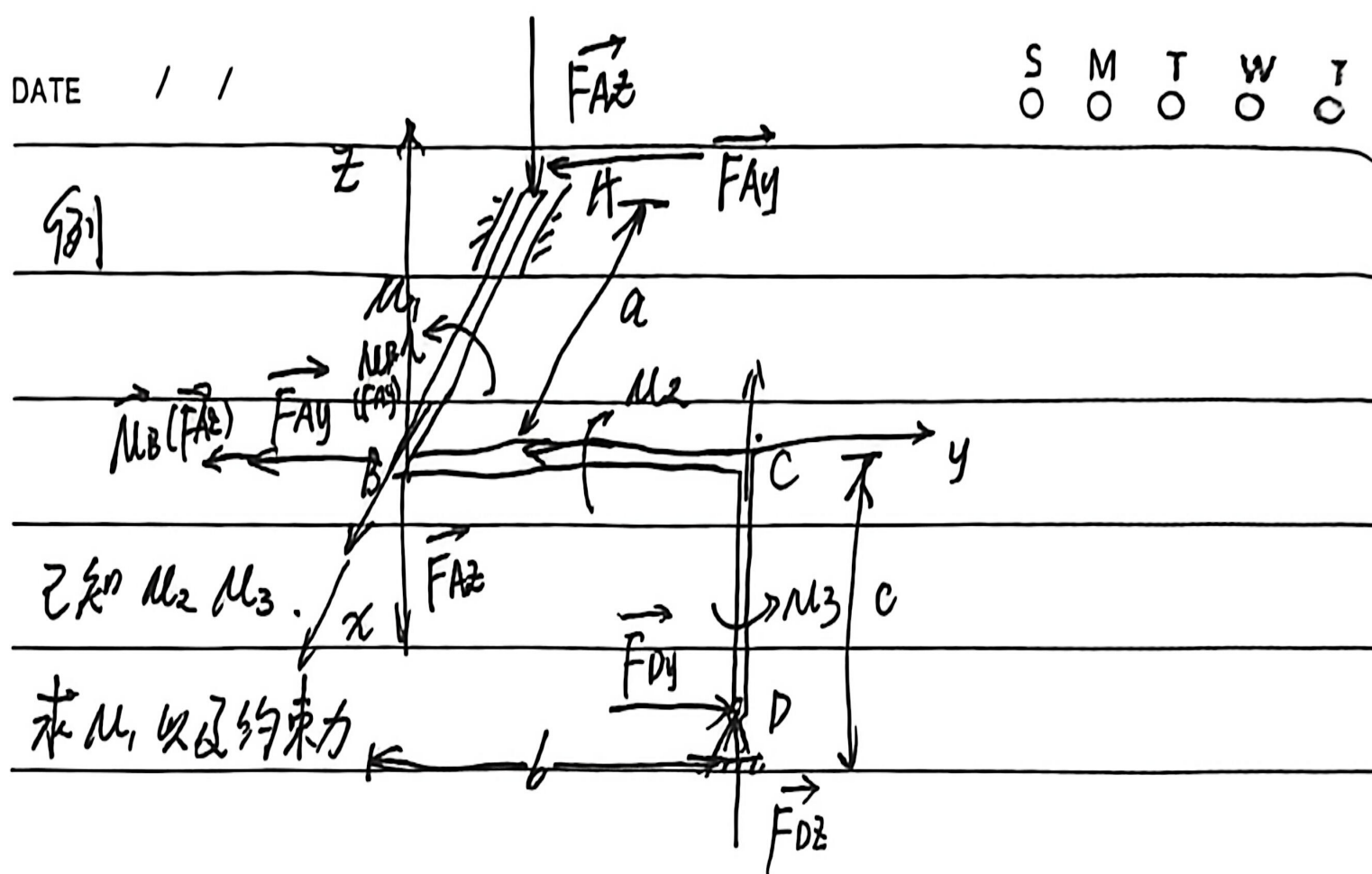


$$F_{Ax} = F_{Bx} \quad F_{Az} = F_{Bz}$$

$$\sum M_{ix} = 0 \quad F_{1z} \cdot 400 - F_{Az} \cdot 800 = 0$$

$$\sum M_{iz} = 0 \quad F_{1x} \cdot 400 - F_{Ax} \cdot 800 = 0$$

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$$F_{Ay} = F_{Dy} \quad F_{Az} = F_{Dz} \quad F_{Ay} \cdot a = N_B(F_{Ay})$$

$$F_{Az} \cdot a = N_B(F_{Az})$$

$$\sum M_{ix} = 0 \quad M_1 + F_{Ay}c + F_{Dz} \cdot b = 0$$

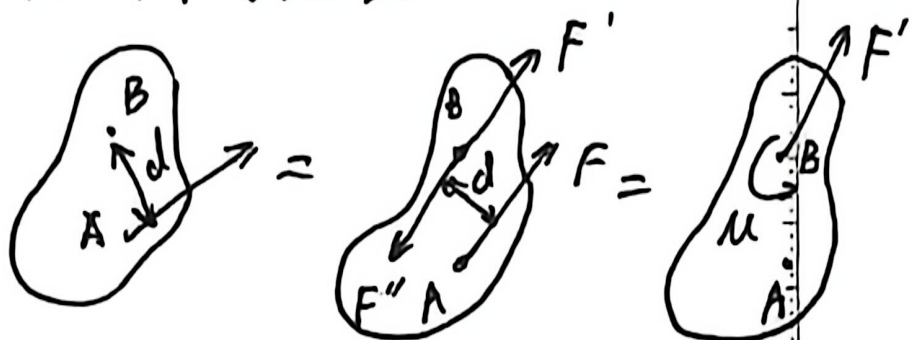
$$\sum M_{iy} = 0 \quad -M_2 - N_B(F_{Az}) = 0$$

$$\sum M_{iz} = 0 \quad M_3 + N_B(F_{Ay}) = 0$$

§2-4 空间任意力系的简化

向一点简化 O 简化中心

一、力的平移定理



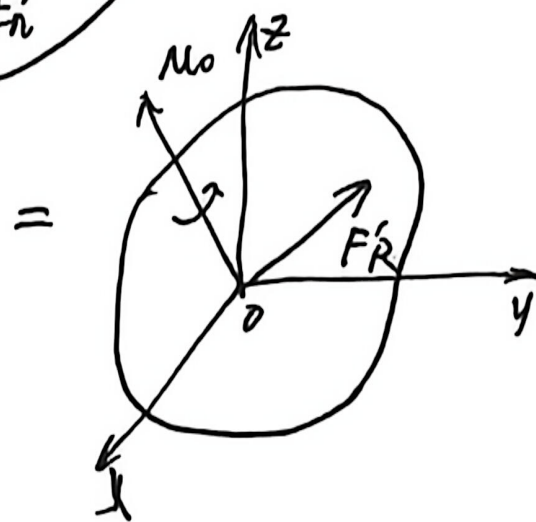
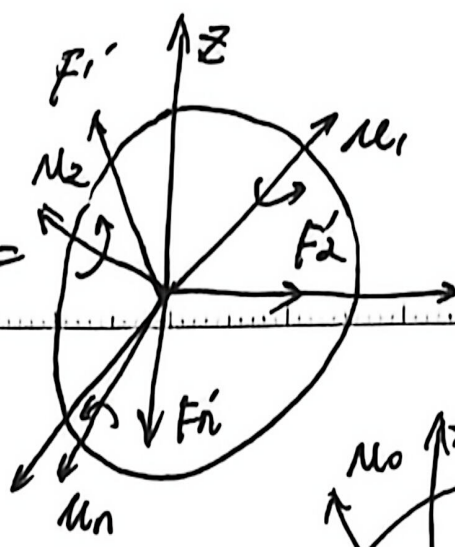
$$\vec{M} = \vec{M}_B(\vec{F})$$

力的平移要附加力偶

二、向一点简化.

$$\begin{cases} \vec{F}_{1i}' = \vec{F}_i \\ \vec{F}_{2i}' = \vec{F}_i \\ \vec{F}_{ni}' = \vec{F}_i \end{cases} \text{ 汇交力系}$$

$$\begin{cases} \vec{M}_i = \vec{M}_O(\vec{F}_i) \\ \vec{M}_2 = \vec{M}_O(\vec{F}_2) \text{ 力偶系} \\ \vec{M}_n = \vec{M}_O(\vec{F}_n) \end{cases}$$



$$\vec{F}_R = \sum \vec{F}_i' = \sum \vec{F}_i \text{ 主矢 不强调作用点与作用线}$$

$$F_R = \sqrt{(\sum F_{ix})^2 + (\sum F_{iy})^2 + (\sum F_{iz})^2}$$

$$\cos(\vec{F}_R, i) = \frac{\sum F_{ix}}{F_R}$$

$$\vec{M}_O = \sum \vec{M}_i = \sum \vec{M}_O(\vec{F}_i) \text{ 主矩}$$

$$M_{Ox} = \sum M_x(\vec{F}_i)$$

$$M_{Oy} = \sum M_y(\vec{F}_i)$$

$$M_{Oz} = \sum M_z(\vec{F}_i)$$

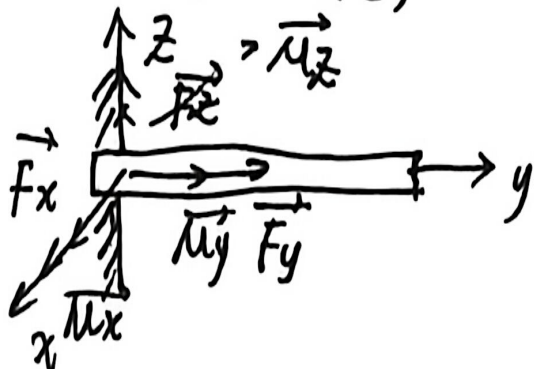
$$M_O = \sqrt{(\sum M_x(\vec{F}_i))^2 + (\sum M_y(\vec{F}_i))^2 + (\sum M_z(\vec{F}_i))^2}$$

$$\cos(\vec{M}_O, i) = \frac{\sum M_x(\vec{F}_i)}{M_O}$$

固定端约束

$$\vec{F}_R = \sum F_{ix} \vec{i} + \sum F_{iy} \vec{j}$$

$$M_o = \sum M_o(\vec{F}_i)$$



空间固定端约束 6个未知量 3个力 3个力偶

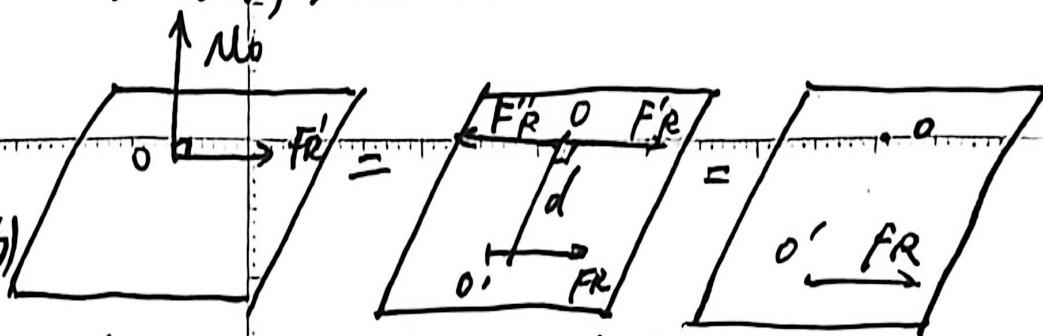
三、简化结果的分析

1. $\vec{F}_R = 0$ $\vec{M}_o \neq 0 \Rightarrow$ 1个力偶, 简化中心可变

2. $\vec{F}_R \neq 0$ $\vec{M}_o = 0 \Rightarrow$ 1个力 (合力) 要通过简化中心

3. $\vec{F}_R \neq 0$ $\vec{M}_o \neq 0$

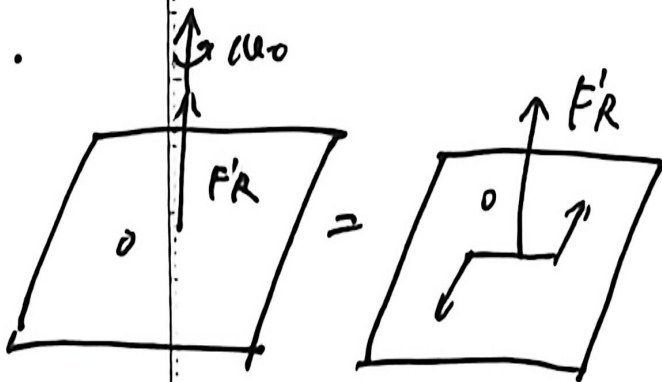
(1) $\vec{F}_R \perp \vec{M}_o \Rightarrow$ 1个力 (合力)



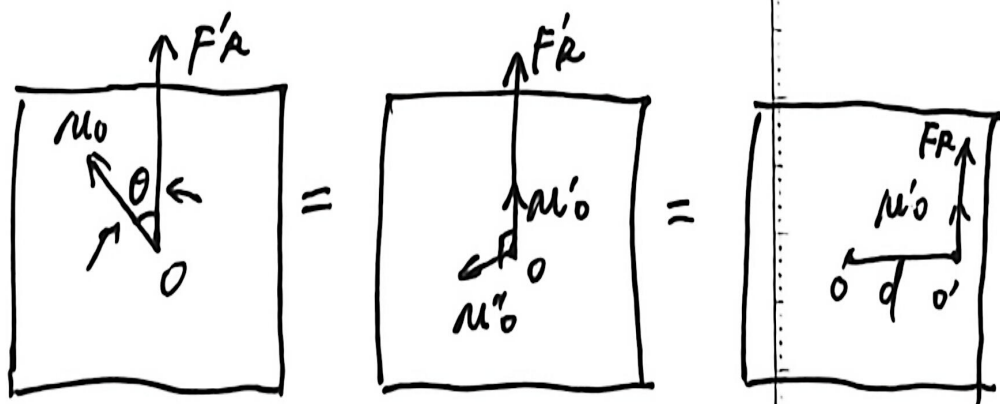
$\vec{M}_o = \vec{M}_o(\vec{F}_R)$ $d = \frac{|\vec{M}_o|}{|\vec{F}_R|}$ 与简化中心有一段距离d.

$= \sum \vec{M}_o(\vec{F}_i)$ 合力矩定理.

(2) $\vec{F}_R \parallel \vec{M}_o$ 力螺旋



(3) $\vec{F}_R$ 与 $\vec{M}_o$ 斜交 力螺旋



4. $\vec{F}_R = 0$ $\vec{M}_o = 0$ 平衡

平面任意力系

a) 一个力偶

(2) 一个力 (合力)

(3) 平衡

2.5 空间任意力系的平衡方程

$$\vec{F}_R = 0 \Rightarrow \sum F_{ix} = 0$$

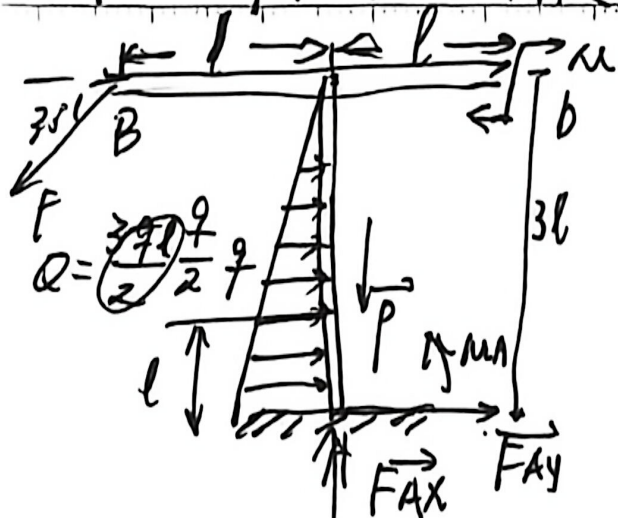
$$\sum F_{iy} = 0$$

$$\sum F_{iz} = 0$$

$$\vec{M}_i = 0 \Rightarrow \sum M_x(\vec{F}_i) = 0$$

$$\sum M_y(\vec{F}_i) = 0$$

$$\sum M_z(\vec{F}_i) = 0$$

例1 已知 P, M, Q, l, F 求A处约束力

固定端约束

$$\sum F_{ix} = 0$$

$$F_{Ax} + Q - F \cos 30^\circ = 0$$

$$F_{Ay} - P - F \sin 30^\circ = 0$$

$$F_{Ay} - P - F \sin 30^\circ = 0$$

取矩心, 使尽量多的未知力通过矩心, $\sum M_o(\vec{F}_i) = 0$, 均等于0则该力系平衡

$$M_A - Q \cdot l - M + F \sin 30^\circ l + F \cos 30^\circ 3l = 0$$

空间平行力系

$$\sum F_{iz} = 0$$

$$\sum M_x(\vec{F}_i) = 0$$

$$\sum M_y(\vec{F}_i) = 0$$

平面任意力系

$$\sum F_{ix} = 0$$

$$\sum F_{iy} = 0$$

$$\sum M_o(\vec{F}_i) = 0$$

$$\sum M_A(\vec{F}_i) = 0$$

$$\sum M_B(\vec{F}_i) = 0$$

$$\sum F_{iz} = 0$$

$$\sum M_A(\vec{F}_i) = 0 \Rightarrow \text{一个过A点的力}$$

$$\sum M_B(\vec{F}_i) = 0 \Rightarrow \text{一个过B点的力}$$

$$F_{AB} = 0$$

$$\sum M_A(\vec{F}_i) = 0$$

ABC不共线

$$\sum M_B(\vec{F}_i) = 0$$

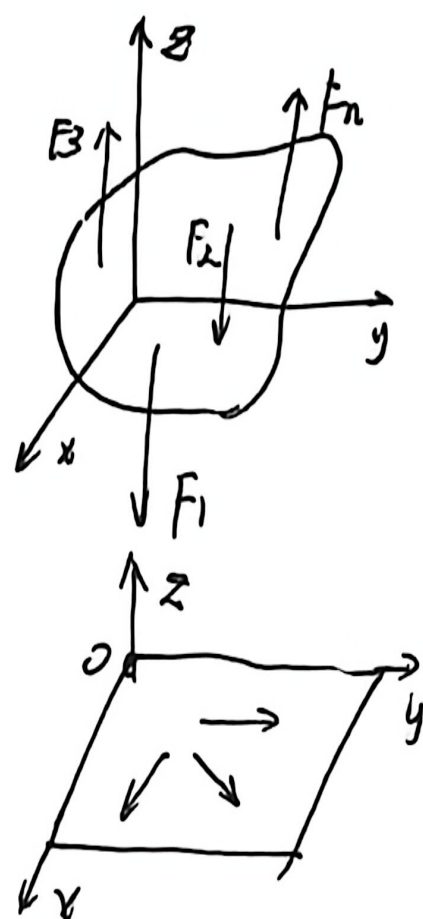
$$\sum M_C(\vec{F}_i) = 0$$

平面平行力系

$$\sum F_{iy} = 0 \quad \text{or} \quad \sum F_{ix} = 0$$

$$\sum M_o(\vec{F}_i) = 0 \quad \sum M_o(\vec{F}_i) = 0$$

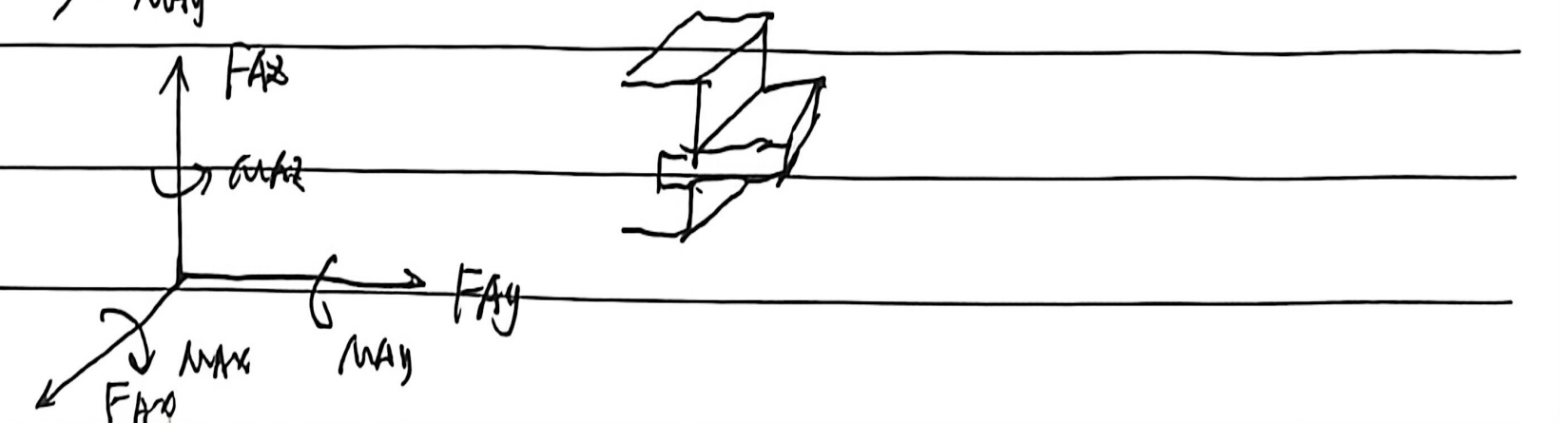
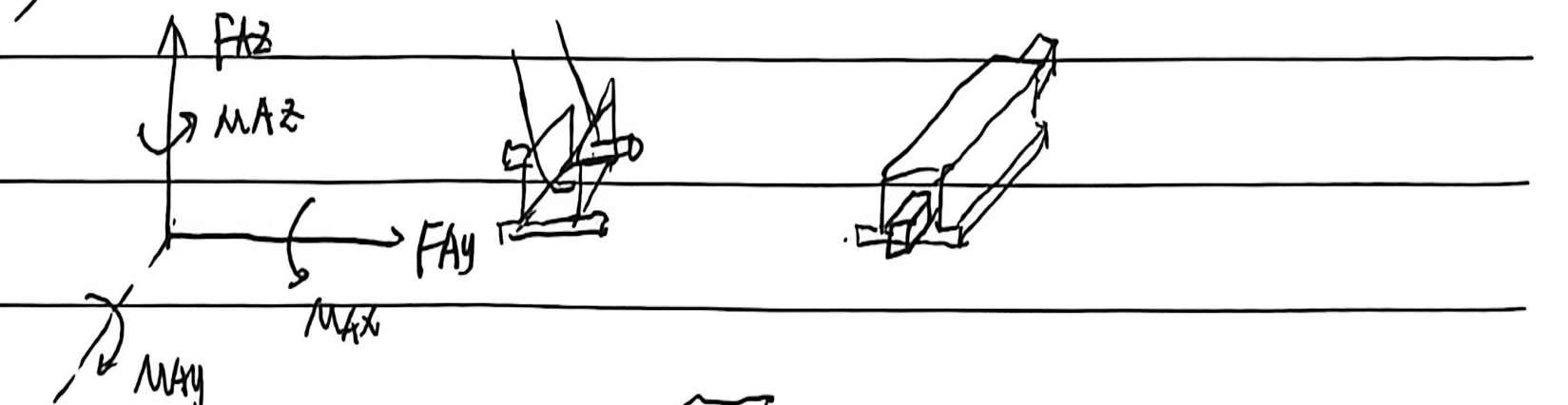
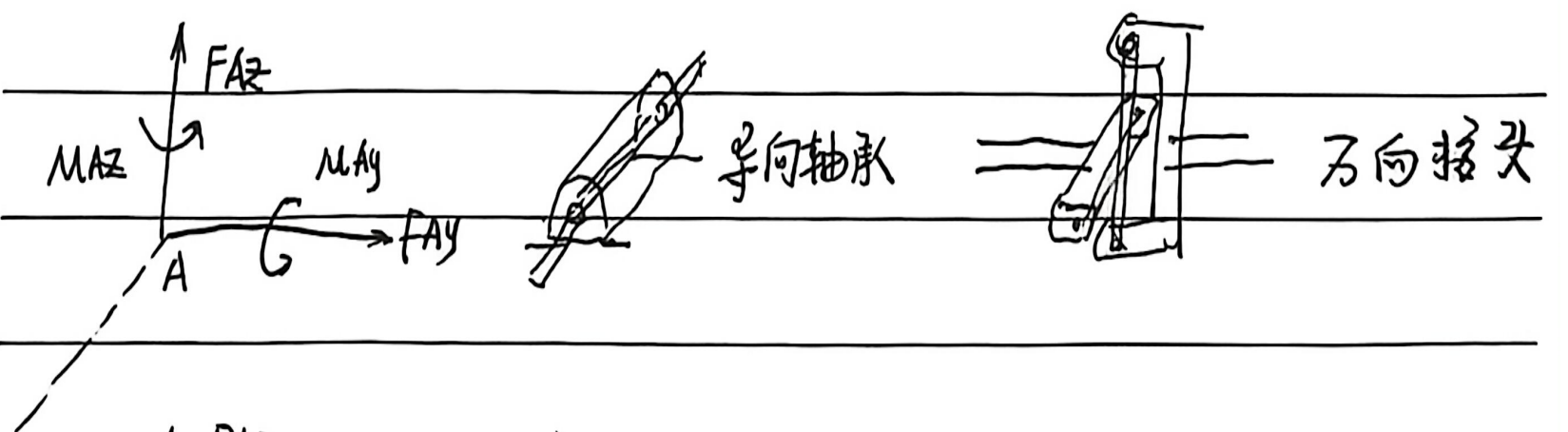
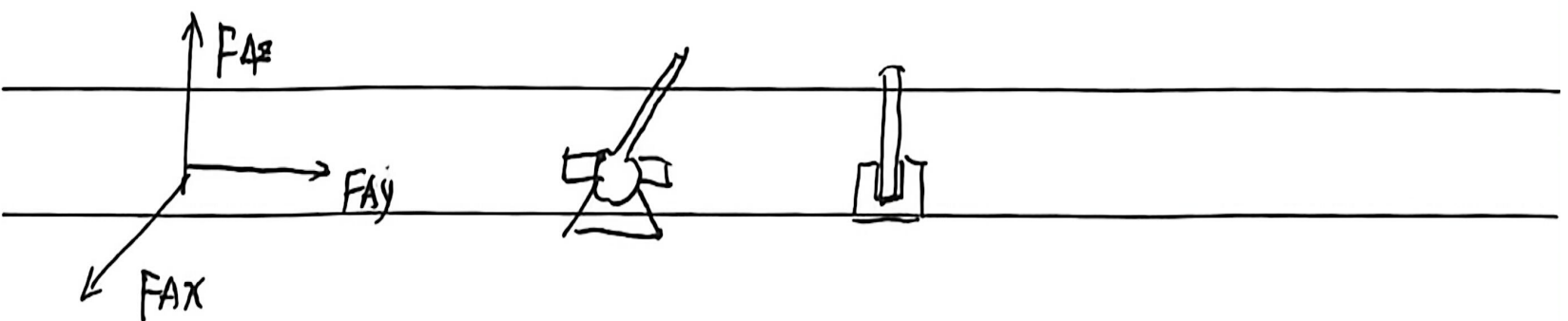
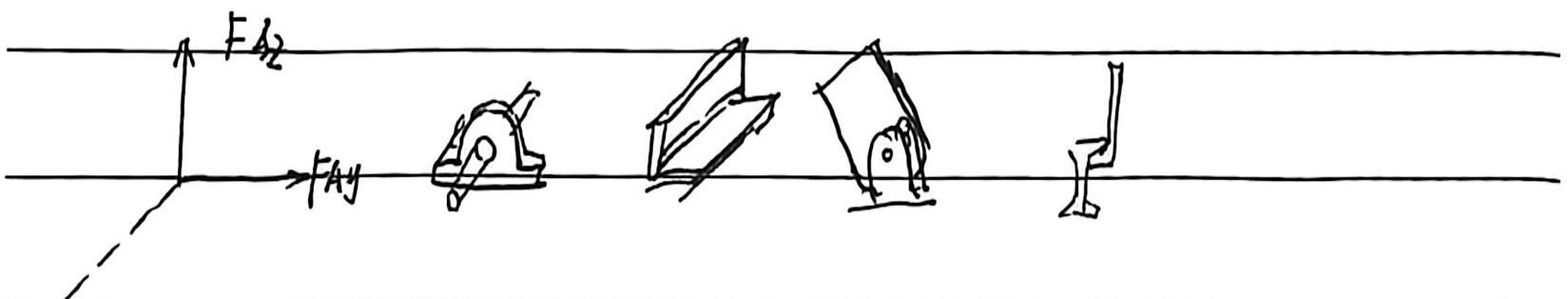
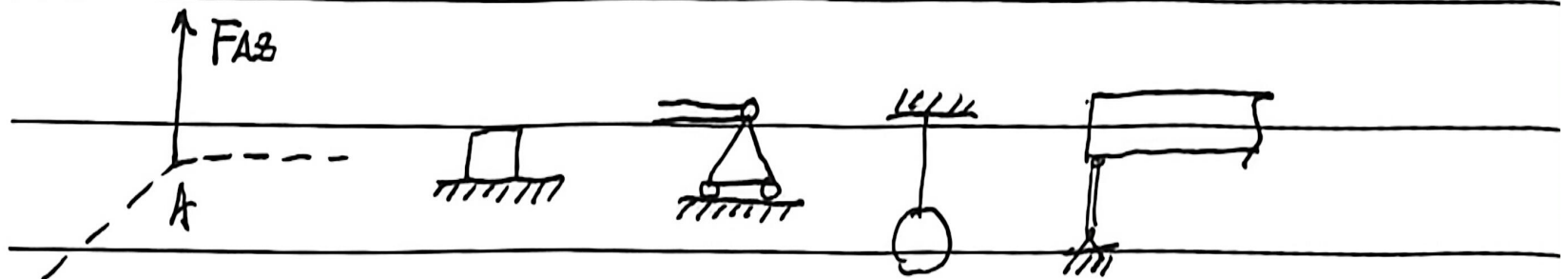
一空间力系若对不共线的任意三点的矩



AB连线不垂直于x轴

各类力系应有的独立平衡方程数

汇交力系	力偶系	平行力系	一般力系
平面 空间	平面 空间	平面 空间	平面 空间
2 3	1 3	2 3	3 6



力偶可以任意移位置只限于同一构件上移动

力可以沿着作用线移动也只限于同一构件上移动

§2-6 物体系统的平衡 静定和超静定问题

平面汇交力系 $n_1 \Rightarrow 2n_1$

平面力偶系 $n_2 \Rightarrow n_2$

平面任意力系 $n_3 \Rightarrow 3n_3$

独立的平衡方程数 $\geq$ 未知力的数目

$2n_1 + n_2 + 3n_3 \Rightarrow$ 静定问题

< 未知力的数目

$\Rightarrow$ 超静定问题 补充方程

超静定次数 = 未知数 - 方程

例1. 已知 q, F, μ, θ 求 A、B 处约束力。

整体 $\sum F_{sx} = 0$

$$F_{Ax} + Q + F \cos \theta - F_B \cos 45^\circ = 0$$

$$\sum F_{sy} = 0$$

$$F_{Ay} - F \sin \theta + F_B \sin 45^\circ = 0$$

$$\sum M(B) = 0$$

$$- \mu + F_{Ax}(b-a) - F_{Ay}(3a) + Q\left(\frac{2}{3}b - a\right) - F \cos \theta a + F \sin \theta 3a = 0$$

